Reinforcement Learning

Adaptive Computation and Machine Learning

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Reinforcement Learning: An Introduction, Richard S. Sutton and Andrew G. Barto

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An Introduction

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In memory of A. Harry Klopf

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Series Foreword

I am pleased to have this book by Richard Sutton and Andrew Barto as one of the first books in the new Adaptive Computation and Machine Learning series. This textbook presents a comprehensive introduction to the exciting field of reinforcement learning. Written by two of the pioneers in this field, it provides students, practitioners, and researchers with an intuitive understanding of the central concepts of reinforcement learning as well as a precise presentation of the underlying mathematics. The book also communicates the excitement of recent practical applications of reinforcement learning and the relationship of reinforcement learning to the core questions in artificial intelligence. Reinforcement learning promises to be an extremely important new technology with immense practical impact and important scientific insights into the organization of intelligent systems.

The goal of building systems that can adapt to their environments and learn from their experience has attracted researchers from many fields, including computer science, engineering, mathematics, physics, neuroscience, and cognitive science. Out of this research has come a wide variety of learning techniques that have the potential to transform many industrial and scientific fields. Recently, several research communities have begun to converge on a common set of issues surrounding supervised, unsupervised, and reinforcement learning problems. The MIT Press series on Adaptive Computation and Machine Learning seeks to unify the many diverse strands of machine learning research and to foster high quality research and innovative applications.

Thomas Dietterich

Preface

We first came to focus on what is now known as reinforcement learning in late 1979. We were both at the University of Massachusetts, working on one of the earliest projects to revive the idea that networks of neuronlike adaptive elements might prove to be a promising approach to artificial adaptive intelligence. The project explored the “heterostatic theory of adaptive systems” developed by A. Harry Klopf. Harry’s work was a rich source of ideas, and we were permitted to explore them critically and compare them with the long history of prior work in adaptive systems. Our task became one of teasing the ideas apart and understanding their relationships and relative importance. This continues today, but in 1979 we came to realize that perhaps the simplest of the ideas, which had long been taken for granted, had received surprisingly little attention from a computational perspective. This was simply the idea of a learning system that *wants* something, that adapts its behavior in order to maximize a special signal from its environment. This was the idea of a “hedonistic” learning system, or, as we would say now, the idea of reinforcement learning.

Like others, we had a sense that reinforcement learning had been thoroughly explored in the early days of cybernetics and artificial intelligence. On closer inspection, though, we found that it had been explored only slightly. While reinforcement learning had clearly motivated some of the earliest computational studies of learning, most of these researchers had gone on to other things, such as pattern classification, supervised learning, and adaptive control, or they had abandoned the study of learning altogether. As a result, the special issues involved in learning how to get something from the environment received relatively little attention. In retrospect, focusing on this idea was the critical step that set this branch of research in motion. Little progress could be made in the computational study of reinforcement learning until it was recognized that such a fundamental idea had not yet been thoroughly explored.

The field has come a long way since then, evolving and maturing in several directions. Reinforcement learning has gradually become one of the most active research areas in machine learning, artificial intelligence, and neural network research. The field has developed strong mathematical foundations and impressive applications. The computational study of reinforcement learning is now a large field, with hundreds of active researchers around the world in diverse disciplines such as psychology, control theory, artificial intelligence, and neuroscience. Particularly important have been the contributions establishing and developing the relationships to the theory of optimal control and dynamic programming. The overall problem of learning from interaction to achieve goals is still far from being solved, but our understanding of it has improved significantly. We can now place component ideas, such as temporal-difference learning, dynamic programming, and function approximation, within a coherent perspective with respect to the overall problem.

Our goal in writing this book was to provide a clear and simple account of the key ideas and algorithms of reinforcement learning. We wanted our treatment to be accessible to readers in all of the related disciplines, but we could not cover all of these perspectives in detail. Our treatment takes almost exclusively the point of view of artificial intelligence and engineering, leaving coverage of connections to psychology, neuroscience, and other fields to others or to another time. We also chose not to produce a rigorous formal treatment of reinforcement learning. We did not reach for the highest possible level of mathematical abstraction and did not rely on a theorem–proof format. We tried to choose a level of mathematical detail that points the mathematically inclined in the right directions without distracting from the simplicity and potential generality of the underlying ideas.

The book consists of three parts. Part I is introductory and problem oriented. We focus on the simplest aspects of reinforcement learning and on its main distinguishing features. One full chapter is devoted to introducing the reinforcement learning problem whose solution we explore in the rest of the book. Part II presents what we see as the three most important elementary solution methods: dynamic programming, simple Monte Carlo methods, and temporal-difference learning. The first of these is a planning method and assumes explicit knowledge of all aspects of a problem, whereas the other two are learning methods. Part III is concerned with generalizing these methods and blending them. Eligibility traces allow unification of Monte Carlo and temporal-difference methods, and function approximation methods such as artificial neural networks extend all the methods so that they can be applied to much larger problems. We bring planning and learning methods together again and relate them to heuristic search. Finally, we summarize our view of the state of reinforcement learning research and briefly present case studies, including some of the most impressive applications of reinforcement learning to date.

This book was designed to be used as a text in a one-semester course, perhaps supplemented by readings from the literature or by a more mathematical text such as the excellent one by Bertsekas and Tsitsiklis (1996). This book can also be used as part of a broader course on machine learning, artificial intelligence, or neural networks. In this case, it may be desirable to cover only a subset of the material. We recommend covering Chapter 1 for a brief overview, Chapter 2 through Section 2.2, Chapter 3 except Sections 3.4, 3.5 and 3.9, and then selecting sections from the remaining chapters according to time and interests. Chapters 4, 5, and 6 build on each other and are best covered in sequence; of these, Chapter 6 is the most important for the subject and for the rest of the book. A course focusing on machine learning or neural networks should cover Chapter 8, and a course focusing on artificial intelligence or planning should cover Chapter 9. Chapter 10 should almost always be covered because it is short and summarizes the overall unified view of reinforcement learning methods developed in the book. Throughout the book, sections that are more difficult and not essential to the rest of the book are marked with a ⋆. These can be omitted on first reading without creating problems later on. Some exercises are marked with a ⋆ to indicate that they are more advanced and not essential to understanding the basic material of the chapter.

The book is largely self-contained. The only mathematical background assumed is familiarity with elementary concepts of probability, such as expectations of random variables. Chapter 8 is substantially easier to digest if the reader has some knowledge of artificial neural networks or some other kind of supervised learning method, but it can be read without prior background. We strongly recommend working the exercises provided throughout the book. Solution manuals are available to instructors. This and other related and timely material is available via the Internet.

At the end of most chapters is a section entitled “Bibliographical and Historical Remarks,” wherein we credit the sources of the ideas presented in that chapter, provide pointers to further reading and ongoing research, and describe relevant historical background. Despite our attempts to make these sections authoritative and complete, we have undoubtedly left out some important prior work. For that we apologize, and welcome corrections and extensions for incorporation into a subsequent edition.

In some sense we have been working toward this book for twenty years, and we have lots of people to thank. First, we thank those who have personally helped us develop the overall view presented in this book: Harry Klopf, for helping us recognize that reinforcement learning needed to be revived; Chris Watkins, Dimitri Bertsekas, John Tsitsiklis, and Paul Werbos, for helping us see the value of the relationships to dynamic programming; John Moore and Jim Kehoe, for insights and inspirations from animal learning theory; Oliver Selfridge, for emphasizing the breadth and importance of adaptation; and, more generally, our colleagues and students who have contributed in countless ways: Ron Williams, Charles Anderson, Satinder Singh, Sridhar Mahadevan, Steve Bradtke, Bob Crites, Peter Dayan, and Leemon Baird. Our view of reinforcement learning has been significantly enriched by discussions with Paul Cohen, Paul Utgoff, Martha Steenstrup, Gerry Tesauro, Mike Jordan, Leslie Kaelbling, Andrew Moore, Chris Atkeson, Tom Mitchell, Nils Nilsson, Stuart Russell, Tom Dietterich, Tom Dean, and Bob Narendra. We thank Michael Littman, Gerry Tesauro, Bob Crites, Satinder Singh, and Wei Zhang for providing specifics of Sections 4.7, 11.1, 11.4, 11.5, and 11.6 respectively. We thank the Air Force Office of Scientific Research, the National Science Foundation, and GTE Laboratories for their long and farsighted support.

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